DENNIS G. PEZAN

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EDUCATION

Bachelor of Science in Operations Research, **United States Military Academy** Expected 2029

- Minor in Artificial Intelligence; Finance Specialization; CQPA: 3.571
- Research Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)
- Conducted AIAD at the Transformation Decision Analysis Center (2026)

RESEARCH EXPERIENCE

Undergraduate Researcher, Department of Mathematical Sciences, USMA 2026 - Present

- Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply, advised by LTC Matthew Webb; extends his MIT Center for Transportation & Logistics doctoral research on reinforcement learning for maximum-flow network interdiction
- Built a 57-node, 79-edge open-source model of Taiwan's crude oil import network (four sources, major straits and canals, three ports) with sourced capacities
- Designed a fixed-budget adversary action set of seven action types, each with a cost and success probability anchored on historical events
- Connected the network to a reinforcement-learning interdiction environment and verified it against an independent max-flow calculation; identified and fixed a targeting-filter issue that could hide the Strait of Hormuz

AI Research Intern, Transformation Decision Analysis Center, Fort Lee, VA 2026

- Academic Individual Advanced Development (AIAD) summer internship under MAJ Erica Mikelson
- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit during engagements
- Retrained agent more than tripled the share of requests filled and outperformed a hand-coded priority-aware baseline across 101 random seeds

Independent Study in Machine Learning, Department of Mathematical Sciences, USMA 2025 - 2026

- Designed and completed an independent study of decision trees, bagging, and random forests (MA289) under LTC Patrick Kuiper, and presented the findings
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer
- Delivered a live AI-assisted coding demonstration of machine-learning workflows to deployed Army officers

PRESENTATIONS

1. **Pezan D** (2026). Reward Design for Reinforcement-Learning Resupply in Logistics Battle Command. *Final Results Brief, Transformation Decision Analysis Center.* Fort Lee, VA, July 22.

PROFESSIONAL DEVELOPMENT

Sea to Shining Sea Domestic AIAD, Defense Technology and Innovation 2026

- Engaged Palantir (Project Maven demonstration), Stanford's Gordian Knot Center (Col. (Ret.) Joseph Felter, Jack Faherty), OpenAI, Apple, and Meta Reality Labs (CTO Andrew Bosworth) on how defense technology is developed and procured
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

PROJECTS

Random Forest Research Site, Independent Study (MA289), USMA 2026
- Interactive explainer comparing decision trees, bagging, and random forests; built and deployed through AI-assisted development (Claude Code, GitHub, Vercel)

CFT Packing List, Web Application 2026
- React web app that cadets use to organize and track gear for the Combat Fitness Test

HONORS AND AWARDS

Dean's List, United States Military Academy Fall 2025, Spring 2026

Stokes Writing Fellow, United States Military Academy

PROFESSIONAL CERTIFICATIONS

Software Engineering Job Simulation, JPMorgan Chase & Co. (Forage) 2026

Modern Data Architecture with Microsoft Fabric, Microsoft (Coursera) 2026

PROFESSIONAL EXPERIENCE

Intern, GrayStone AI Solutions, Remote 2024
- Built Python and Excel tools to support financial planning and analysis workflows

Department Manager, McDonald's, Plainfield, IL 2022 - 2024
- Managed weekly scheduling for 80+ employees with 95% on-time accuracy and oversaw $20,000+ in weekly supply procurement
- Promoted as the youngest department manager in the region; sustained 90%+ customer satisfaction scores

LEADERSHIP AND VOLUNTEER SERVICE

Cadet Ambassador, CLD-STEM Program, USMA 2025 - Present
- Represent West Point at advisory boards in Chicago and Los Angeles; delivered 50+ hours of STEM outreach and mentorship to underrepresented students

Certified Tutor, Center for Enhanced Performance, USMA 2025 - Present
- Tutor cadets in mathematics, chemistry, and psychology

Member, Finance Club, USMA 2025 - Present
- Engage with industry executives from companies including Palantir, Uber, and Meta

Volunteer Senior Publicist, Recycle & Reboot, Plainfield, IL 2024
- Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

Candidate, Illinois Premier Boys State 2024
- Elected Alderman and Township Supervisor; advanced to the Lieutenant Governor primary through competitive public speaking

PROGRAMMING LANGUAGES AND SKILLS

Python, Java (Spring Boot), SQL, JavaScript (React)
Microsoft Fabric, Kafka, REST APIs, Git/GitHub, Claude Code
Spanish (fluent) ◇ Security Clearance: Secret